

ESAS

The Estimation System ©

LOBACHEVSKY UNIVERSITY

ESYS – The Evaluation System App
Overview and Features Walkthrough

Denis Khorkin
Arzamas Branch of Lobachevsky State University of Nizhny Novgorod
2023

Contents

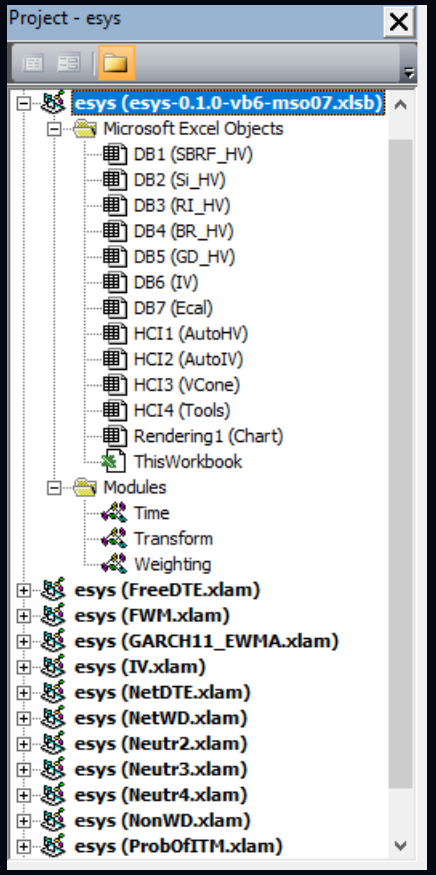
- Intro
- Architecture
- Data
- Market Conditions
- Historical Volatility (HV) Interface
- Implied Volatility (IV) Interface
- Volatility Cone (VC) Interface
- Conclusion
- References

The initial prototype was released on July 6, 2016. It is called **The Evaluation System (ESYS)** and is written in VBA. **VBA (Visual Basic for Applications)** is a programming language developed by Microsoft. Excel, along with the other members of Microsoft Office, includes the VBA language (at no additional cost). Simply put, VBA is the tool that people use to develop programs that control Excel. VBA is a programming language, but it also serves as a macro language. In this context, a macro is a set of instructions Excel performs to imitate keystrokes and mouse actions [1]. VBA code is written in the Visual Basic Editor (VBE) within Microsoft Office applications.

The prototype consists of class modules, modules, and add-ins:

- **A class module** is designed to create custom objects. It provides a template for defining the structure and behavior of objects.
- **A VBA module** is a container for VBA code and can contain procedures, functions, and variables. It is primarily used for storing general-purpose code.
- **VBA add-ins** extend the capabilities of Microsoft Excel. An Excel add-in has the .xlam file extension and is a workbook that Excel can open automatically when it starts up.

Prototype: Architecture



Class Modules

- SBRF_HV–GD_HV: databases of settlement price and historical volatility (HV) for 100-, 50-, 20-, and 10-day periods of underlying assets;
- IV: implied volatility (IV) database;
- Ecal: options expiration calendar;
- AutoHV, AutoIV: UI for automatic calculation of HV and IV;
- VCone: UI for automatic calculation of the volatility cone;
- Tools: tools for working with futures and options;
- Chart: rendering of 2D charts;
- ThisWorkbook: the workbook containing the currently running code.

Modules

- Time: time handling;
- Transform: data transformation;
- Weighting: data weighting.

Add-ins

- FWM: Fishburne weighting method;
- FreeDTE, NetDTE, NetWD, NonWD: HV normalizers;
- Neutr2, Neutr3, Neutr4: risk neutralizers for 2, 3, and 4 options;
- GARCH11_EWMA, IV, ProbOfITM: not covered here (require real-time data).

LOBACHEVSKY UNIVERSITY

Moex File Specification for Futures

FUTREPORT = YYYY-TRADABLESYMBOL-TLD.EXTENSION
where
 YYYY: settlement year, 4 symbols (0000-9999)
 TRADABLESYMBOL = CMY
 where
  C: code of the underlying asset, 2 symbols
  M: settlement month (for options it also
     determines contract type - call or put),
     1 symbol (Jan=F, Feb=G, Mar=H, Apr=J,
     May=K, Jun=M, Jul=N, Aug=Q, Sep=U, Oct=V,
     Nov=X, Dec=Z)
  Y: settlement year, 1 symbol (0-9)
 TLD: top-level domain (ru)
 EXTENSION: file extension (csv)
Example of a report name:
2012-RIZ2-ru.csv, *2022-RIZ2-ru.csv*
The leading year (YYYY) is added to eliminate confusion associated with the ten-year cycle.

Moex File Specification for Options

OPTREPORT = YYYYMMDD-FILENAME-TLD.EXTENSION
where
 YYYYMMDD: complete calendar date. For more
           information, see ISO 8601
 FILENAME: moex file name (rts-results,
           pr-options-delta)
 TLD: top-level domain (ru)
 EXTENSION: file extension (csv)
Example of a report name:
20220103-rts-results-ru.csv,
20220103-pr-options-delta-ru.csv

Prototype: Data (continued)

LOBACHEVSKY UNIVERSITY

Description of fields in CSV for Futures traded on Moex
(for example, for *2022-RIZ2-ru.csv*)
Date|Average Weighted Price for the lot|Settlement Price|First Trade|High|Low|Last Trade|Change,%|Last Trade Volume, contracts|Number of trades|Total Volume, RUR|Total Volume, Contracts|Open Interest, RUR|Open Interest, Contracts|

Description of fields in CSV for Options traded on Moex
(for example, for *20220103-rts-results-ru.csv*)
Contract code|Delivery|Weighted average price for the lot|Settlement Price|First trade|Max|Min|Last trade|Change|Last trade volume (contracts)|Trades num|"Trades Volume, RUR"|Trades Volume|"Open Interest, RUR"|"Open Interest, Contracts"|

(for example, for *20220103-pr-options-delta-ru.csv* (published in Russian only))
Дата|Код опциона|Право по опциону|Последний торговый день, в который может быть заключен опционный контракт|Цена исполнения опционного контракта|Количество подлежащих исполнению опционных контрактов|Сумма подлежащих исполнению опционных контрактов|Коэффициент дельта|Теоретическая цена опциона|Волатильность|

Trade results
https://www.moex.com/en/derivatives/contracts.aspx
https://www.moex.com/ru/derivatives/contracts.aspx
The values of the *"Trades Volume"* field
https://www.moex.com/en/archive/fortsmarketresults.aspx
https://www.moex.com/ru/archive/fortsmarketresults.aspx
The values of the *"Волатильность"* field
https://www.moex.com/ru/forts/coefficients-values.aspx

LOBACHEVSKY UNIVERSITY

Asset Price Dynamics

Motion Model of Asset Prices

For concreteness, consider options on futures and denote the current futures price by F. F represents the current price of the underlying asset. The asset price during a trading session is assumed to follow a continuous random process, with no jumps between sessions. For simplicity, the price is described as a discrete process with time step τ: $F_0 = F, F_1, F_2, \ldots, F_i$. The step can be one day, one hour, one minute, etc., and is expressed in fractions of a year. Since the process exists only during trading sessions, there are typically 252 trading days in a calendar year, 63 trading days in a quarter, and 21 trading days in a month. For example, if the step is one day, then $\tau = 1/252$.

A discrete-time model for prices, modeled as geometric Brownian motion, is given by

$$(F_i - F_{i-1})/F_{i-1} = \mu\tau + \sigma\sqrt{\tau}Z_k$$

where

$(F_i - F_{i-1})/F_{i-1}$: the relative change

μ: the average rate of the price trend, expressed as a simple annual percentage

σ: volatility

$Z_1, Z_2, \ldots, Z_k$: independent Gaussian random variables with zero mean and a standard deviation of 1 (from a normal distribution)

Prototype: Market Conditions (continued)

				WMA(100)			W100	0,1	WMA(50)			W50	0,2	WMA(20)			W20	0,3	WMA(10)			W10	0,4			
TradeDate	Settle	Ui	Ui²	δ²n	δ	σ	Ū	μ	δ²n	δ	σ	Ū	μ	δ²n	δ	σ	Ū	μ	δ²n	δ	σ	Ū	μ	δ²n comp	σ comp	σ transpos
16.03.2023	76609	0,010073	0,000101	0,000107	0,010368	0,16459	0,002414	0,608387	6,85E-05	0,008278	0,131408	0,001364	0,343822	7,37E-05	0,008586	0,1363	0,002567	0,646892	5,83E-05	0,007636	0,121221	0,002635	0,664351	6,47E-05	0,126509	
17.03.2023	76853	0,003192	1,02E-05	0,000105	0,01027	0,163031	0,002418	0,609351	6,59E-05	0,00812	0,128895	0,00125	0,314916	6,66E-05	0,008158	0,129512	0,00232	0,584523	5,2E-05	0,00721	0,114461	0,002399	0,604482	6,99E-05	0,132119	
20.03.2023	77247	0,005127	2,63E-05	0,000104	0,010186	0,161704	0,002338	0,589257	6,4E-05	0,008001	0,127014	0,001254	0,316019	6,12E-05	0,007823	0,124188	0,002052	0,517164	4,9E-05	0,006998	0,111091	0,002587	0,65191	6,45E-05	0,12672	
21.03.2023	76825	-0,00546	2,98E-05	0,000102	0,010107	0,160445	0,002408	0,606847	6,22E-05	0,00789	0,125244	0,001291	0,325373	5,66E-05	0,007522	0,119404	0,002352	0,592672	4,63E-05	0,006806	0,108037	0,002247	0,566121	6,11E-05	0,123266	
22.03.2023	77175	0,004556	2,08E-05	0,0001	0,010019	0,159052	0,002319	0,584334	6,01E-05	0,007755	0,123103	0,001897	0,478041	5,16E-05	0,007181	0,113999	0,002109	0,531374	4,15E-05	0,006445	0,102318	0,00195	0,4864	5,82E-05	0,120129	
23.03.2023	75986	-0,01541	0,000237	0,000103	0,010146	0,161064	0,002139	0,539039	6,7E-05	0,008187	0,129962	0,001571	0,395947	6,75E-05	0,008217	0,130439	0,001302	0,328062	7,69E-05	0,008767	0,13917	0,000477	0,120288	5,42E-05	0,115653	
24.03.2023	77053	0,014042	0,000197	0,000105	0,01023	0,1624	0,002345	0,590846	7,22E-05	0,008494	0,13484	0,002176	0,548377	7,85E-05	0,008861	0,140656	0,001683	0,424217	0,000101	0,010028	0,159194	0,001764	0,444001	7,47E-05	0,136899	
27.03.2023	76845	-0,0027	7,29E-06	0,000103	0,010128	0,160781	0,002336	0,588772	6,99E-05	0,00836	0,132703	0,002362	0,595131	7,07E-05	0,008407	0,133452	0,000878	0,221233	8,62E-05	0,009284	0,147376	0,002662	0,670929	8,87E-05	0,149083	
28.03.2023	77125	0,003644	1,33E-05	0,000101	0,010031	0,159239	0,002267	0,571261	6,8E-05	0,008243	0,13086	0,002165	0,545625	6,42E-05	0,008014	0,127222	0,002254	0,5681	7,53E-05	0,008675	0,137715	0,002553	0,668486	7,99E-05	0,141605	
29.03.2023	77378	0,00328	1,08E-05	9,86E-05	0,009931	0,157658	0,002297	0,578834	6,61E-05	0,008128	0,129023	0,002302	0,580045	6,02E-05	0,007759	0,123166	0,002161	0,544523	6,39E-05	0,007993	0,126882	0,002034	0,51268	7,3E-05	0,135348	
30.03.2023	77445	0,000866	7,5E-07	9,64E-05	0,009821	0,155898	0,002437	0,614187	6,38E-05	0,007986	0,126772	0,002119	0,533913	5,53E-05	0,007436	0,118038	0,001875	0,472503	5,21E-05	0,00722	0,114608	0,001114	0,280654	6,67E-05	0,129273	
31.03.2023	77787	0,004416	1,95E-05	9,47E-05	0,009729	0,15445	0,002485	0,626307	6,23E-05	0,007893	0,1253	0,002376	0,598724	5,24E-05	0,007237	0,114878	0,001817	0,457997	4,56E-05	0,006753	0,107199	0,001234	0,311511	5,98E-05	0,122199	
03.04.2023																								5,59E-05	0,117848	

Four types of volatility

1. Future ("true") volatility (FV), also known as actual volatility – cannot be measured directly;
2. Historical volatility (HV) – calculated directly from historical prices;
3. Forecast volatility – estimated from data (HV extrapolation: mean, GARCH, EWMA) or predicted by a model (e.g., GBM assuming Gaussian returns, volatility cone, implied volatility);
4. Implied volatility (IV) – derived from options' market prices (one method).

Historical volatility is derived from the prices of the underlying asset. To estimate the parameters μ and σ for a discrete set of prices $F_0 = F, F_1, F_2, \ldots, F_i$, it is necessary to determine the relative price changes over period τ:

- Settle: settlement price of the futures contract;
- U_i: relative change per period τ, $U_i = (F_i - F_{i-1})/F_{i-1}$;
- U_i^2: square of relative change (used to estimate volatility and variance of the trend; always non-negative);
- Ū: average trend rate (mean relative change per period), $\bar{U} = (1/m)\Sigma U_{n-i}$ (captures the direction and average rate of the trend).

Parameter Estimation for Period τ (continued)

- δ^2_n: Fishburne weighted variance, $\delta^2_n = \Sigma \alpha_i (U_{n-i})^2$, where $\Sigma \alpha_i = 1$;
- δ: weighted volatility estimate for period τ (not annualized), $\delta = \sqrt{\delta^2_n}$. δ is the standard deviation of random variables $U_1, U_2, \ldots, U_i$.

Estimates of the coefficients μ and σ are obtained by normalization:

- μ: the average rate of the price trend (expressed as a simple annual percentage), $\mu = \bar{U}/\tau$;
- σ: annualized volatility estimate (δ normalized by $\sqrt{\tau}$), $\sigma = \delta/\sqrt{\tau}$, where $\tau = 1/252$.

Window Principle

- Count window: groups data streams according to the number of elements;
- Sliding count window: calculates the sum of the most recent 100 elements every 10 elements;
- **Fishburne formula:** $\alpha_i = 2(n - i + 1)/(n(n + 1))$.

Prototype: HV Interface

HOLS	WSAT	Param	Value	Weight
02.01.2023		Unit	1	
23.02.2023		FNetWH	0,6042	
08.03.2023		FNonWH	0,3958	
01.05.2023		NetWD	254	
09.05.2023		NetWHWD	153,4668	1
12.06.2023		NonWHWD	100,5332	1
		NonWD	0	0
		CompLOT	254	
		ExpDate	19.04.2023	
		NetDTE	12	
		NetWHDTE	7,2504	1
		NonWHDTE	4,7496	1
		FreeDTE	0	0
		CompDTE	12	

Directory	C:\Users\0mega3\Desktop\esys\data\buf\autohv\SBRF\
JoinMode	AM
JoinDepth	10
Estimator	Vyz
Frequency	1
LOT	254
DTE	18
BegDate	24.03.2023
EndDate	24.03.2023

TradeDate	Settle	Open	High	Low	Close	VarDTE	VolaDTE	Var100	Vola100	Var50	Vola50	Var20	Vola20	Var10	Vola10

Trading Schedule

Main trading session:
09:00–10:00 = 60 min;
10:00–14:00 = 240 min;

Intraday clearing session:
14:00–14:05 = 5 min;

Main trading session:
14:05–18:50 = 285 min;

Evening clearing session:
18:50–19:05 = 15 min;

Evening trading session:
19:05–23:50 = 285 min;

Unit: unit multiplier (1 = 1 day)
- 1D to H: 1–24 (Unit x 24);
- 1D to M: 1–1440 (Unit x 1440);
- 1D to S: 1–86400 (Unit x 86400).

FNetWH: Fraction of Net Working Hours
- Net trading time (NetTT) = 870 min;
- FNetWH = NetTT/1440 = 0.6042.

FNonWH: Fraction of Non-Working Hours
- Non-trading time (NonTT) = 570 min;
- FNonWH = NonTT/1440 = 0.3958.

NetWD: Net Workdays
The number of whole working days between start_date and end_date. Working days exclude weekends and any dates identified in holidays (HOLS). It is also adjusted for working Saturdays (WSAT).

NetWHWD: Net Working Hours of Workdays
NetWHWD = NetWD · FNetWH · Weight

NonWHWD: Non-Working Hours of Workdays
NonWHWD = NetWD · FNonWH · Weight

NonWD: Non-Working Days
The length of the Gregorian year (LGY) consists of 365 days (common year), with a leap day being added to February in the leap years.
NonWD = LGY · Unit – (NetWHWD + NonWHWD)

CompLOT: Composite Length of Time
CompLOT = NetWHWD + NonWHWD + NonWD

ExpDate: Option Expiration

NetDTE: Net Days to Expiration

NetWHDTE: Net Working Hours of Days to Exp
NetWHDTE = NetDTE · FNetWH · Weight

NonWHDTE: Non-Working Hours of Days to Exp
NonWHDTE = NetDTE · FNonWH · Weight

FreeDTE: Free Days to Expiration
The number of time intervals (NTI) between two specified dates.
FreeDTE = NTI · Unit – (NetWHDTE + NonWHDTE)

CompDTE: Composite Days to Expiration
CompDTE = NetWHDTE + NonWHDTE + FreeDTE

Prototype: HV Interface (continued)

Directory: the path to market data.
JoinMode: the join mode for selected fields from the prices of expired and continuing contracts.
Price gaps on contract rollover are adjusted to avoid incorrect decisions.
The fields used to calculate the adjustment default to Settlement and OHLC prices.
JoinMode parameters: AM (arithmetic mean, default), GM (geometric mean).
JoinDepth: the join depth for expired and continuing contracts.
It is determined by the number of trading days before expiry.
Rollover trigger:
- If JoinDepth = 1, rollover strictly on the expiry day of the contract;
- If JoinDepth = 0, rollover strictly on the day after the expiry day;
- If JoinDepth > 1, rollover a number of days before the expiry day.

Estimator: computational methods of historical volatility (HV).
Abbr: prices taken (PT), handle drift (HD), handle overnight jumps (HOJ), efficiency (E).
Estimator parameters:
- Vcc – Close-to-Close (PT: C; HD: no; HOJ: no; E(max): 1);
- Vp – Parkinson (PT: HL; HD: no; HOJ: no; E(max): 5.2);
- Vgk – Garman-Klass (PT: OHLC; HD: no; HOJ: no; E(max): 7.4);
- Vrs – Rogers-Satchell (PT: OHLC; HD: yes; HOJ: no; E(max): 8);
- Vgkyz – Garman-Klass Yang-Zhang ext. (PT: OHLC; HD: no; HOJ: yes; E(max): 8);
- Vyz – Yang-Zhang (PT: OHLC; HD: yes; HOJ: yes; E(max): 14).

If the “Estimator” field is empty, prices are displayed.

Frequency: frequency of measurement (e.g. minutely/hourly/daily/weekly/monthly).
LOT: length of time (e.g. number of minutes/hours/days/weeks/months).
DTE: days to expiration.
If the “DTE” field is empty, it is calculated:
- Var100: 100-day variance;
- Vola100: 100-day volatility;
- Var50: 50-day variance;
- Vola50: 50-day volatility;
- Var20: 20-day variance;
- Vola20: 20-day volatility;
- Var10: 10-day variance;
- Vola10: 10-day volatility.

BegDate: begin date.
EndDate: end date.
If the “BegDate” and “EndDate” fields are empty, the workspace is cleared.
VarDTE: DTE-day variance.
VolaDTE: DTE-day volatility.
Settle: settlement price.

Prototype: IV Interface

Directory	C:\Users\0mega3\Desktop\esys\data\buf\autoiv\										
ProdCode	SBRF										
Lifespan	M										
Combo	1		FrontAggIV_Call	FrontAggIV_Put	FrontAvgAggIV	NextAggIV_Call	NextAggIV_Put	NextAvgAggIV	BackAggIV_Call	BackAggIV_Put	BackAvgAggIV	ComboAvgAggIV
Percentage	25	TradeDate										

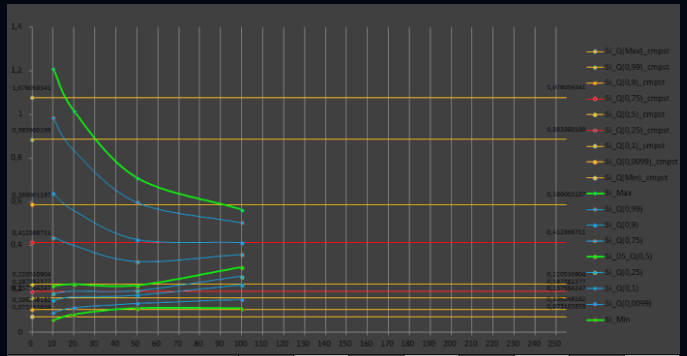
Directory: the path to market data.
ProdCode: the underlying asset code, also known as product code.
If the "ProdCode" field is empty, the workspace is cleared.
Lifespan: option lifespan.
All options have a limited lifespan – Weekly (W), Monthly (M), Quarterly (Q) – and every option contract is defined by an expiration month.
Combo: option series combinator.

- If the "Combo" field is empty, it is set to 1 by default;
- If "Combo" = 1, there is no combo;
- If "Combo" = 2, front IV and next IV are combined;
- If "Combo" = 3, front IV, next IV, and back IV are combined.

Percentage: the percentage (pct) is the strike range: ± pct% of the ATM (at-the-money) strike.
If the "Percentage" field is empty, it is set to 0% by default.
Percentage applies only to strikes with zero volume.
The aggregated IV (agg) – using the Fishburne formula – consists of ATM IV, weighted IV by non-zero volume, IV with zero volumes above the CS (central strike), and IV with zero volumes below the CS.

FrontAggIV_Call: the front agg IV of the call option.
FrontAggIV_Put: the front agg IV of the put option.
FrontAvgAggIV: the average between the front agg IV of the call option and the front agg IV of the put option.
A front month is the nearest expiration date for a futures or options contract.
NextAggIV_Call,
NextAggIV_Put,
NextAvgAggIV:
the next month is the expiration month for a futures or options contract in the following month.
BackAggIV_Call,
BackAggIV_Put,
BackAvgAggIV:
back-month contracts have later expiration dates than front-month contracts and are also called far-month contracts.
ComboAvgAggIV: the average agg IV across the combined series (front, next, back), depending on the "Combo" parameter.

Prototype: VC Interface



	Si					Depth_HV	2	Depth_IV	2	Lifespan	M	Combo	1
HalfPipe	Lip	Vert	Transition			Flat		Transition				Vert	Lip
Decile	Min	D1	D2	D3	D4	D5	D6	D7	D8			D9	Max
IV	0,125914	0,135996	0,143214	0,149534	0,153673	0,157814	0,160175	0,164537	0,171673			0,181547	0,212387
HV_10d	0,057529	0,099291	0,122453	0,143379	0,173415	0,21423	0,270748	0,34139	0,427024			0,584222	1,21054
HV_20d	0,083371	0,112328	0,13279	0,149606	0,174775	0,22117	0,268968	0,394875	0,476236			0,61211	1,014549
HV_50d	0,110414	0,132609	0,154661	0,17943	0,198431	0,215696	0,323112	0,469964	0,514436			0,604575	0,708156
HV_100d	0,110649	0,144022	0,17016	0,185765	0,207099	0,299657	0,411357	0,505634	0,519754			0,533109	0,562011
Quantile(P)	0,0013	0,0099	0,1	0,25		0,5		0,75	0,9			0,99	0,9987
Z-score	-3	-2,3264	-1,2816	-0,6745		0		0,6745	1,2816			2,3264	3
Q(P)IV		0,133076	0,144186	0,150642				0,170084	0,181127			0,200134	
Q(P)HV_10d		0,092714	0,147287	0,178999				0,438234	0,639854			0,986835	
Q(P)HV_20d		0,114312	0,162302	0,190188				0,399548	0,560101			0,836409	
Q(P)HV_50d		0,134053	0,170719	0,192025				0,326417	0,426075			0,597582	
Q(P)HV_100d		0,153088	0,218913	0,257162				0,358643	0,411735			0,503104	
ΣW=1	0,5	0,4	0,06	0,04									
Q(p)HV_cmpst	0,073164	0,106248	0,157564	0,187382		0,220511		0,412867	0,586001			0,88396	1,076059

Volatility Cone Parameters

DepthHV: depth of HV data (expressed in years).

DepthIV: depth of IV data (expressed in years).

Lifespan: option lifespan.

Combo: option series combinator.

Volatility Cone Matrix

HalfPipe	Lip	Vert	Transition			Flat		Transition		Vert	Lip
Decile	Min	D1	D2	D3	D4	D5	D6	D7	D8	D9	Max
IV											
HV											
Quantile	0.0013	0.0099	0.1	0.25		0.5		0.75	0.9	0.99	0.9987
Z-score	-3.0	-2.3264	-1.2816	-0.6745		0		0.6745	1.2816	2.3264	3.0

HalfPipe: the character of a halfpipe (hp) depends on the relationship between four attributes: lip (upper and lower limits), vert, transition, flat (the middle line). Each attribute is assigned a set of deciles.

Decile: a decile is a quantitative method of splitting up a set of ranked data into 10 equally large subsections [2].

Quantile(P): inverse value of the standard normal distribution function.

In statistics and probability, quantiles are cut points dividing the range of a probability distribution into continuous intervals with equal probabilities, or dividing the observations in a sample in the same way. There is one fewer quantile than the number of groups created [3].

Z-score: the z-score of a normal distribution.

Z-score, or standard score, is the number of standard deviations a given data point lies above or below the mean. The mean is the average of all values in a group, added together, and then divided by the total number of items in the group [4].

StdDev to Z-score mapping

StdDev: −3σ; −2σ; −1σ; 0; +1σ; +2σ; +3σ.

Z-score: −3.0; −2.0; −1.0; 0; +1.0; +2.0; +3.0.

General formula for calculating quantile values by Z-score (Z):

D5Value + (D5Value − MinValue) / 3 · (−Z)

D5Value + (MaxValue − D5Value) / 3 · (+Z)

LOBACHEVSKY UNIVERSITY

Why options?

Directional trading

Directional trading refers to strategies based on the investor's view of the future course of something: either the overall financial market or a particular security. Their assessment of the direction will be the sole determining factor in whether the investor decides to sell or buy [5].

- Directional trading with strictly fixed risk (buying call and put options);
- Directional trading, like trading an underlying asset, using synthetic futures (buying and selling call and put options).

Hedging

The best way to understand hedging is to think of it as a form of insurance. When people decide to hedge, they are insuring themselves against a negative event's impact on their finances. This doesn't prevent all negative events from happening. However, if a negative event does happen and you're properly hedged, the impact of the event is reduced [6].

Market making

Making a market is an action whereby a dealer or market maker stands by, ready to make a two-sided quote. This quote indicates they are willing and able to either buy or sell a particular security at the quoted bid and ask price [7].

Trading volatility

- Trading implied volatility (kappa/vega risk);
- Replication of options (delta-gamma risk).

References

[1] Alexander M., Walkenbach J. Excel VBA Programming for Dummies. – John Wiley & Sons, 2018.

[2] Hayes A. (2021, January 19). What Is a Decile? Definition, Formula to Calculate, and Example. Retrieved 14:20, April 5, 2023, from https://www.investopedia.com/terms/d/decile.asp

[3] Wikipedia contributors. (2023, January 16). Quantile. In Wikipedia, The Free Encyclopedia. Retrieved 14:23, April 5, 2023, from https://en.wikipedia.org/w/index.php?title=Quantile&oldid=1134085249

[4] "The Investopedia Team". (2021, October 01). Z-Score vs. Standard Deviation: What's the Difference? Retrieved 14:27, April 5, 2023, from https://www.investopedia.com/ask/answers/021115/what-difference-between-standard-deviation-and-z-score.asp

[5] Ganti, A. (2022, May 23). Directional Trading. Retrieved 14:35, April 5, 2023, from https://www.investopedia.com/terms/d/directionaltrading.asp

[6] "The Investopedia Team". (2022, April 06). Beginner's Guide to Hedging: Definition and Example of Hedges in Finance. Retrieved 14:45, April 5, 2023, from https://www.investopedia.com/trading/hedging-beginners-guide/

[7] Hayes A. (2022, April 29). Make a Market. Retrieved 15:01, April 5, 2023, from https://www.investopedia.com/terms/m/makeamarket.asp

LOBACHEVSKY UNIVERSITY

Thank You



https://vk.ru/thequanttrader